ePortfolio TM126: Activity 11.2.3 configure DHCP on a Wireless Router	
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Introduction:

This activity demonstrates how to implement a DHCP configuration. The objective is for the client to connect to a wireless router via the DHCP server, which must assign an automatic IP address and verify network connectivity.

Objectives:

1. Connect three computers to a wireless router.

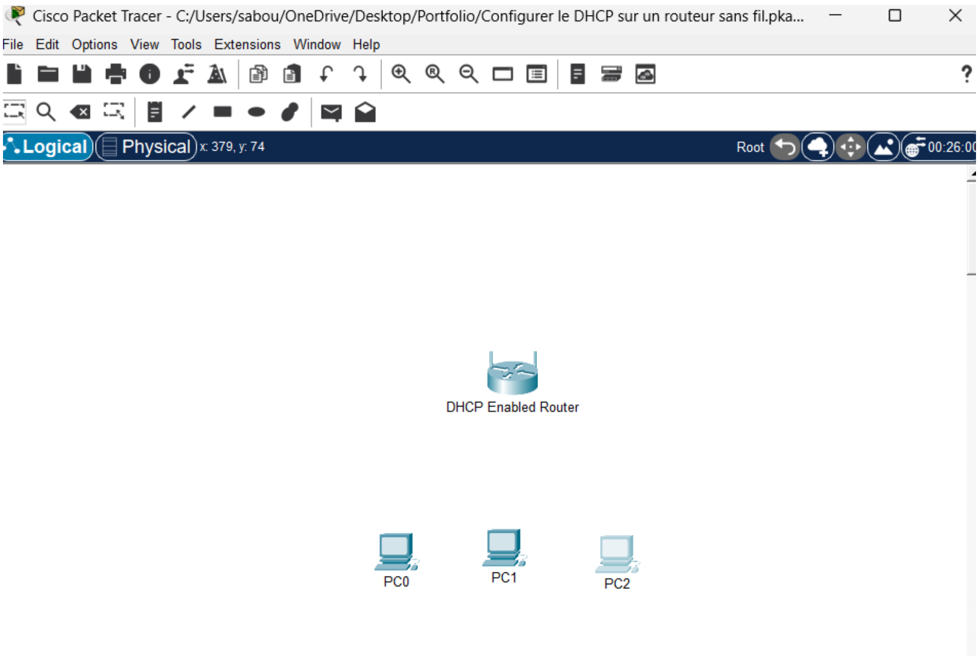
2. Set the DHCP parameter to a specific network range.

3. Configure clients to obtain their address via DHCP.

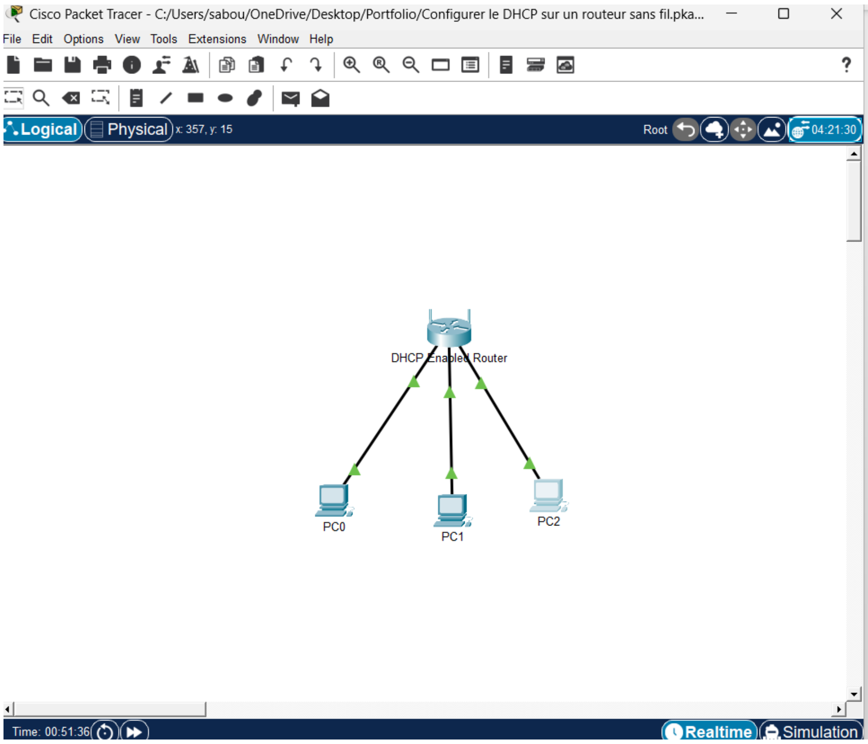
Instructions

Part 1: Set up the network topology

a) Add three generic PCs.



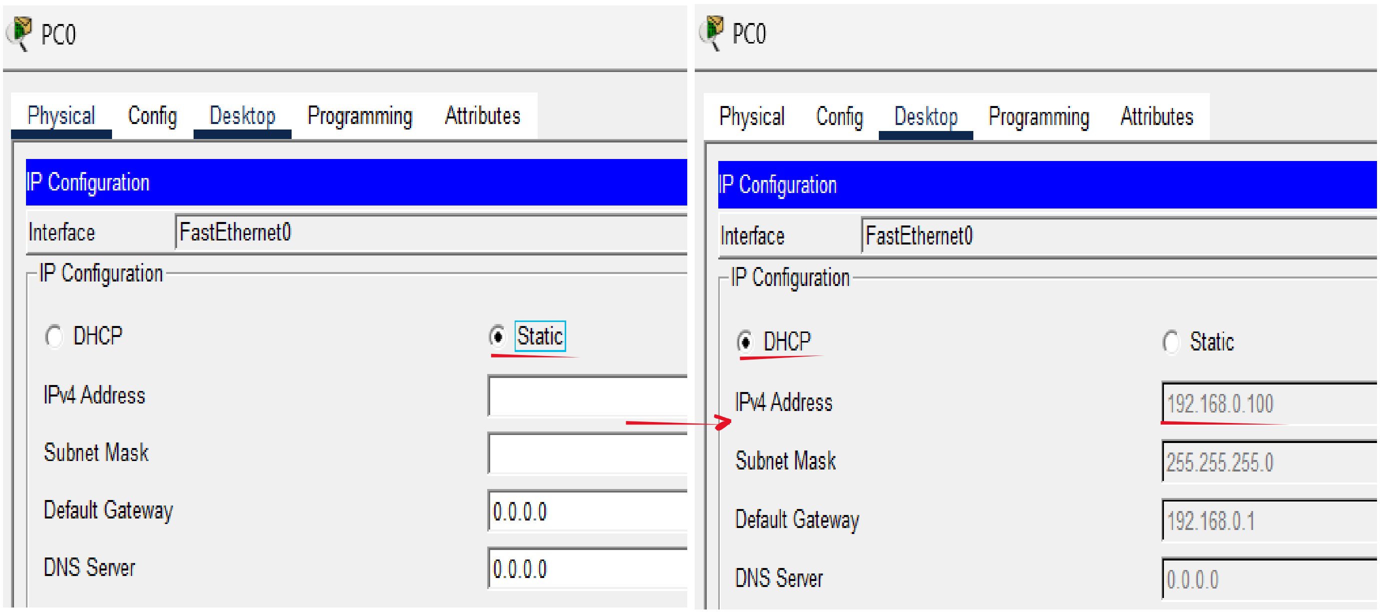
b) Connect each PC to an Ethernet port to the wireless router using straight-through cables.



Part 2: Observe the default DHCP settings

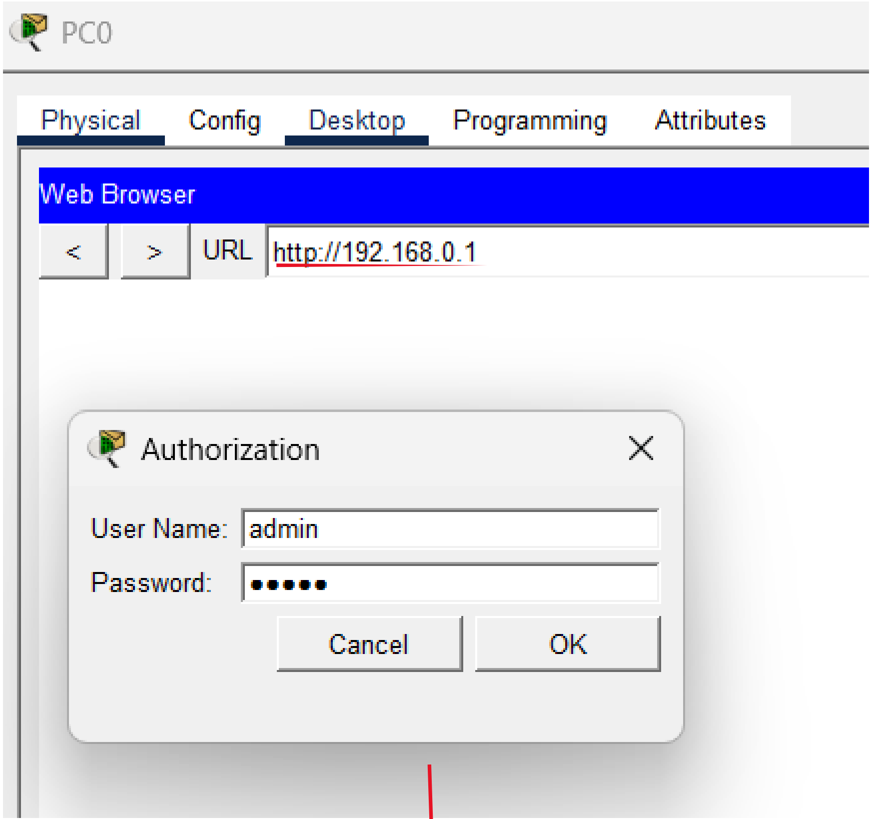
a) Verification of IP addresses obtained automatically on PC0 via the DHCP server

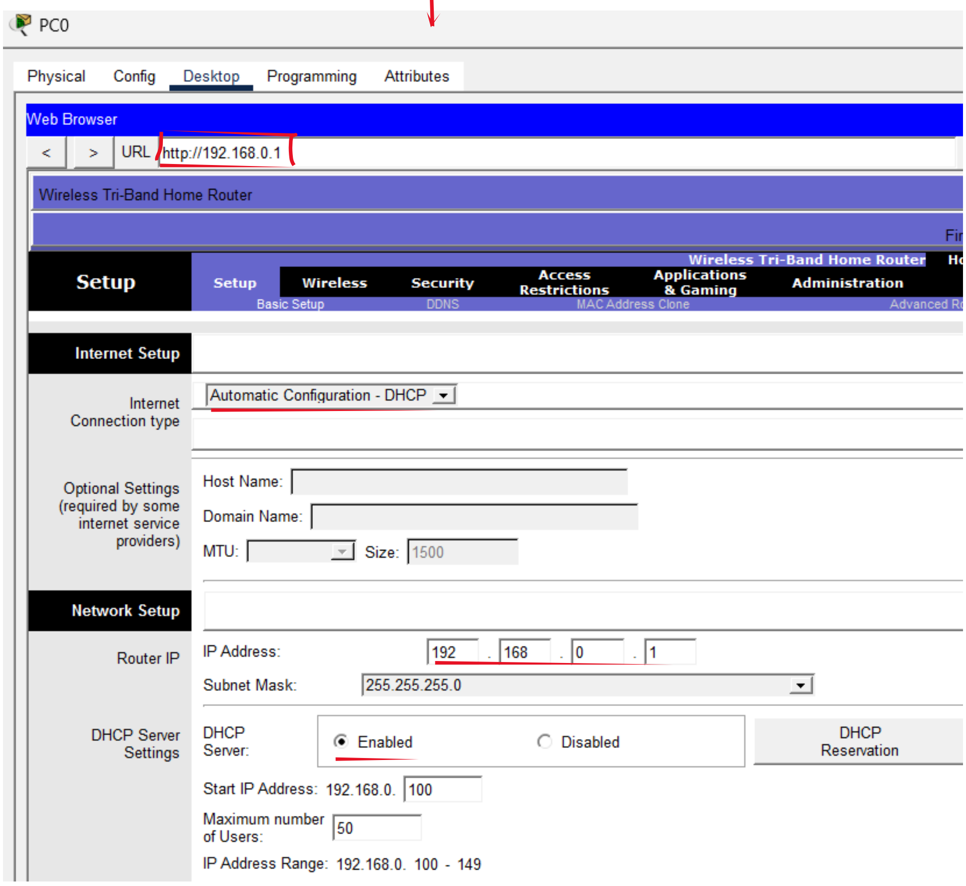
PC0 has obtained an automatic IP address.



The IP address of the default gateway: is **192.168.0.1**

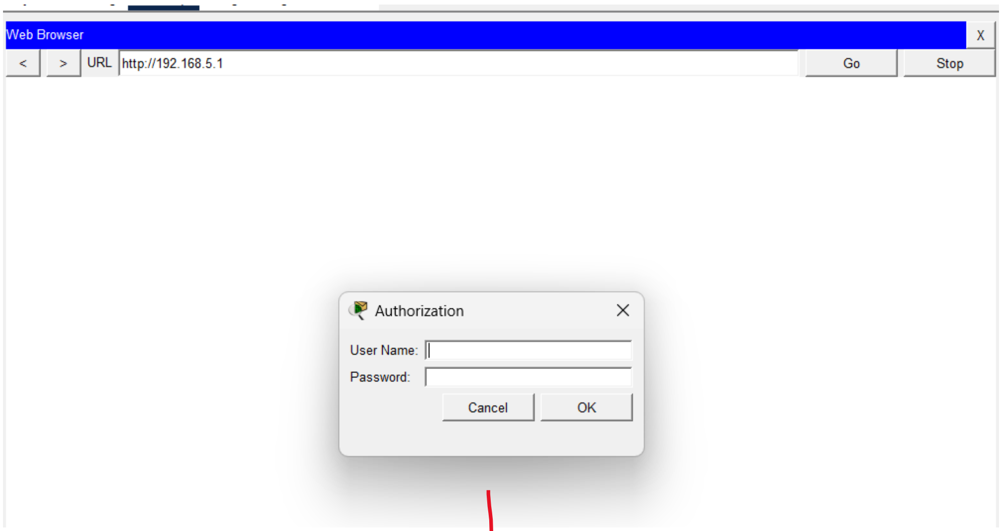
b) Access the router's administration panel check The DHCP server and its setting

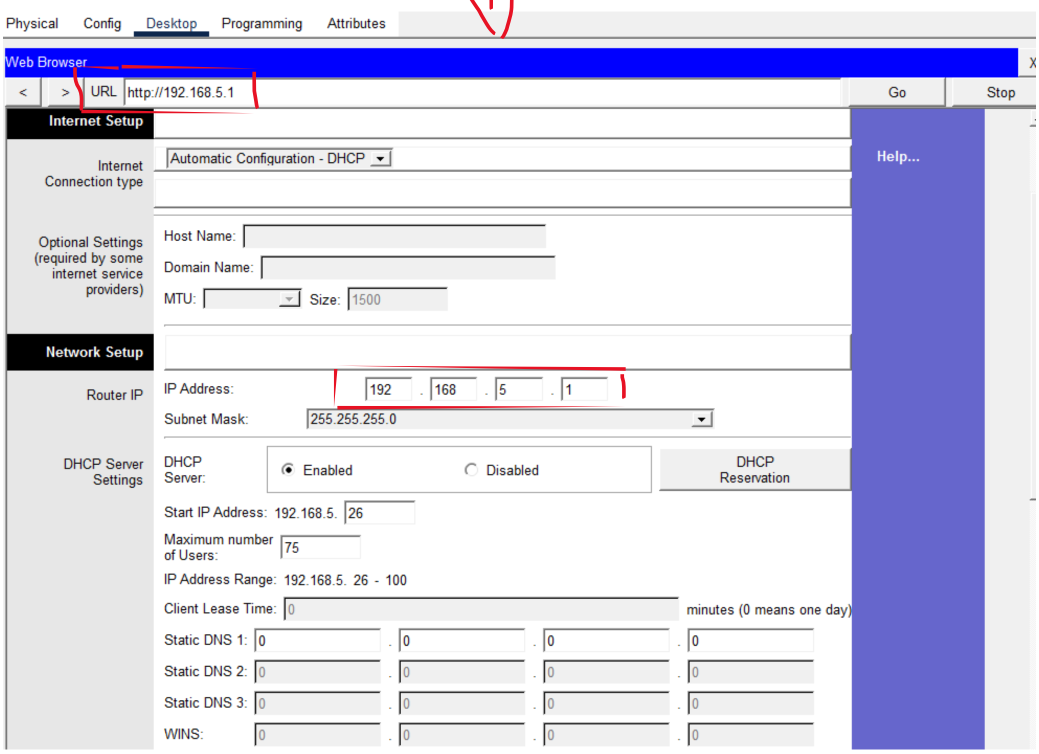




Part 3: Change the default IP address of the wireless router.

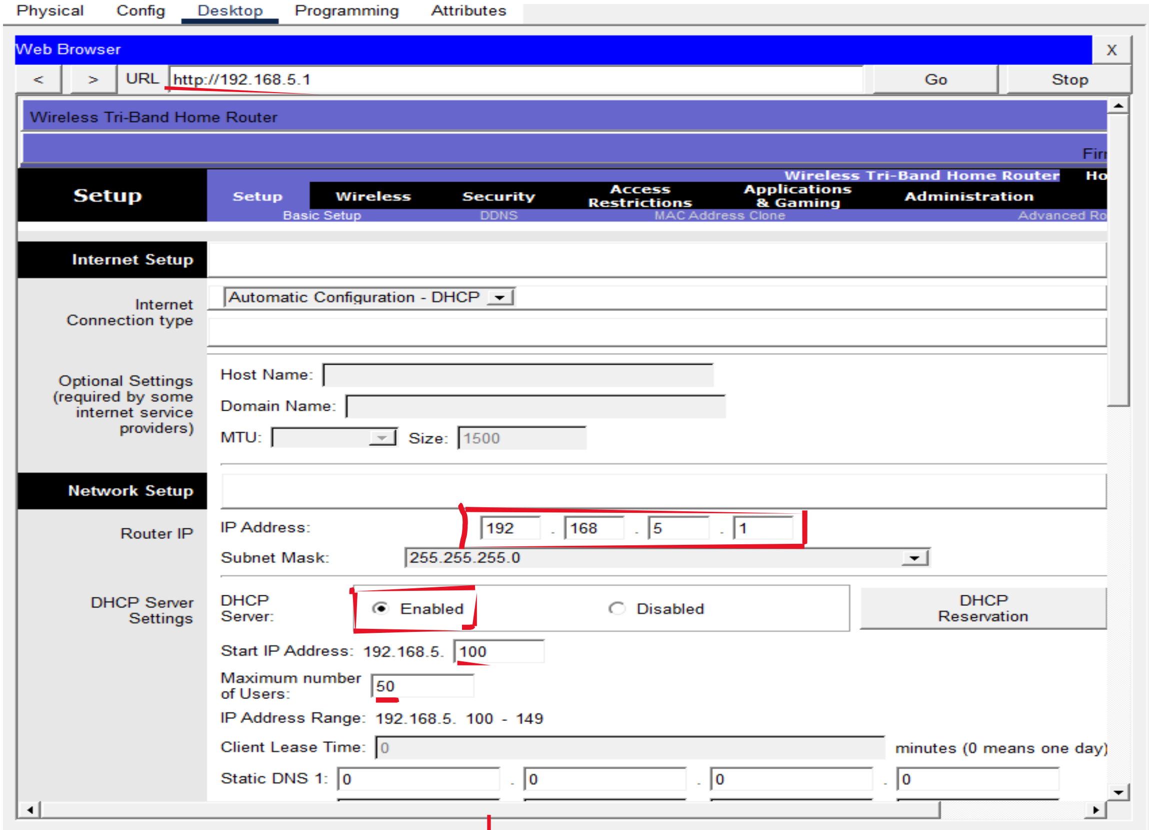
3a) Changed the router IP address from 192.168.0.1 to 192.168.5.1

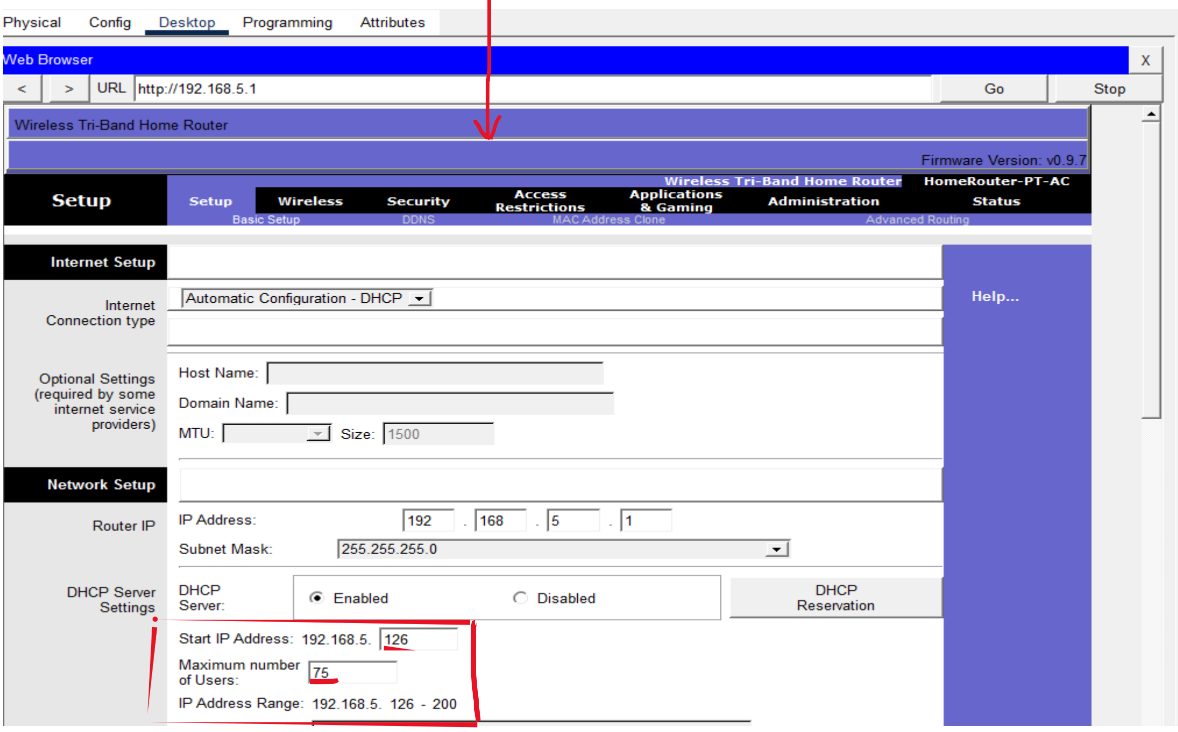




Part 4: Change the default DHCP range of addresses.

Defining an IP address range from 192.168.5.100 to **192.168.5126** for the DHCP server and Maximum Number of Users to **75**.





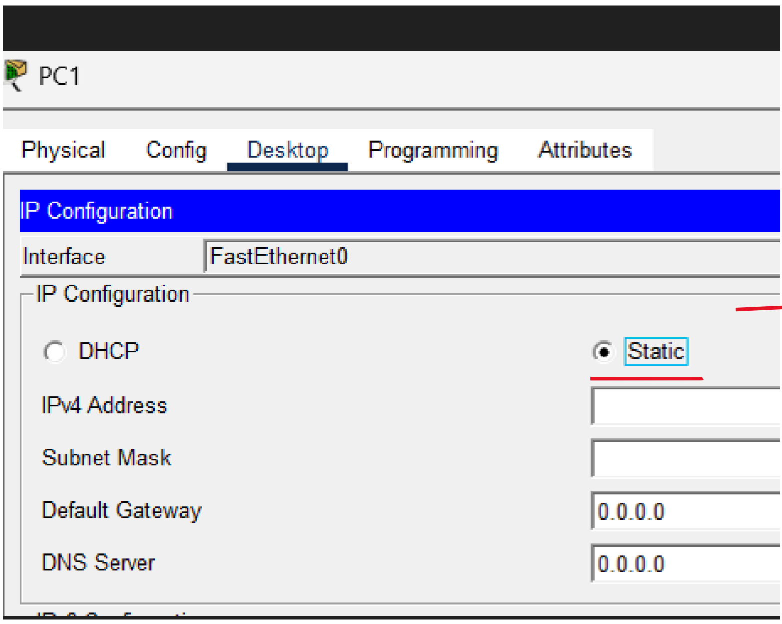
Record the IP address for PC0: the IP address for PC0 is **192.168.1.126**

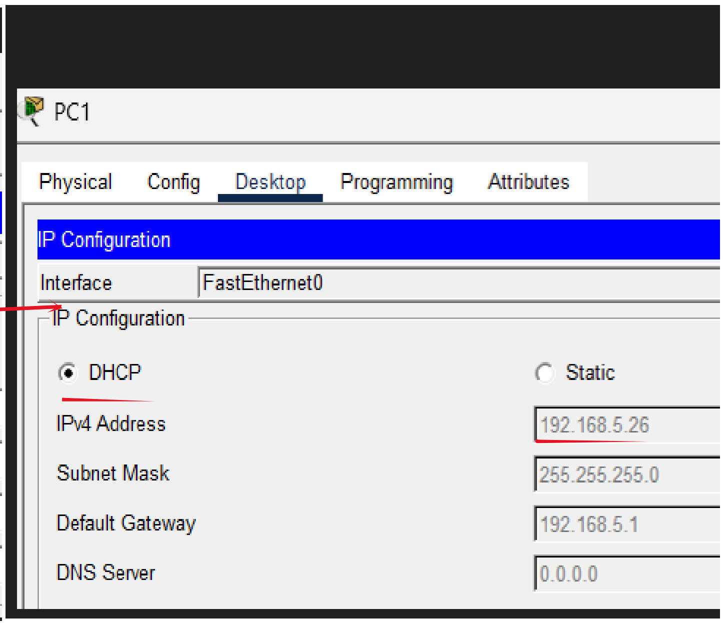
Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>
C:\>ipconfig

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: FE80::2D0:BCFF:FE68:6DE7
   IPv6 Address....................: ::
   IPv4 Address....................: 192.168.5.126
   Subnet Mask.....................: 255.255.255.0
   Default Gateway.................: ::
                                      192.168.5.1
```

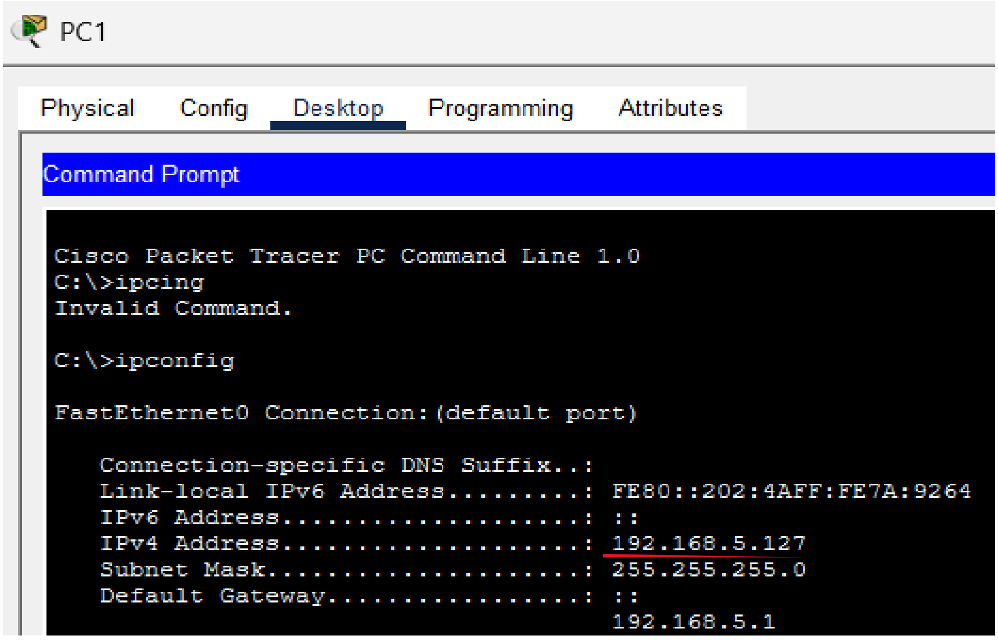
Part 5: Enable DHCP on the other PC1

Testing the automatic acquisition of new IP addresses on PC1

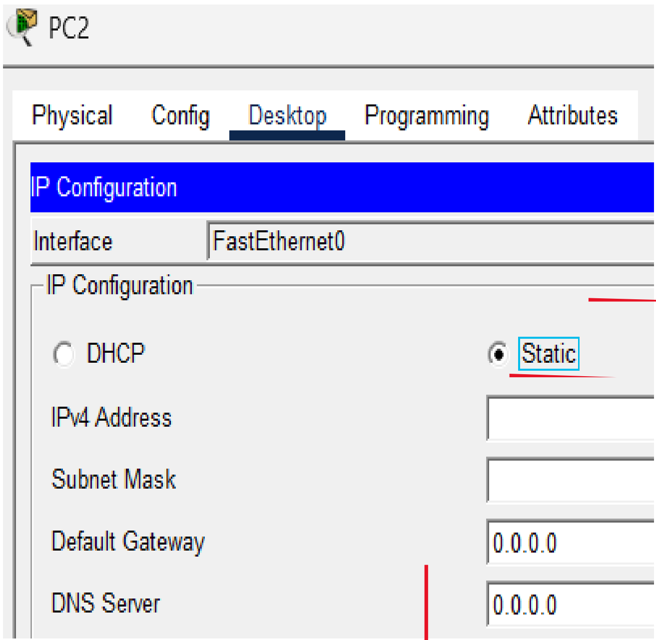


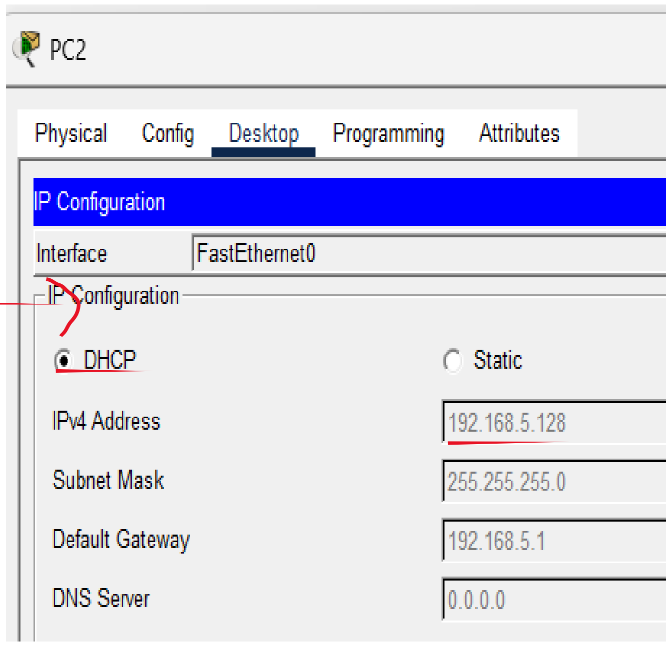


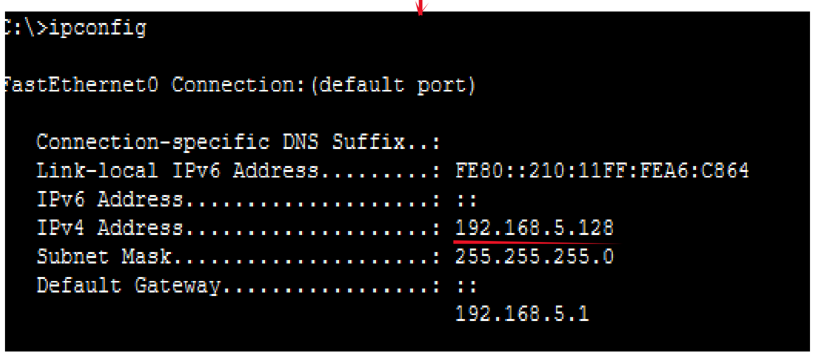
Record the IP address for PC1: the IP address for PC1 is **192.168.1.127**



Enable DHCP on **PC2**



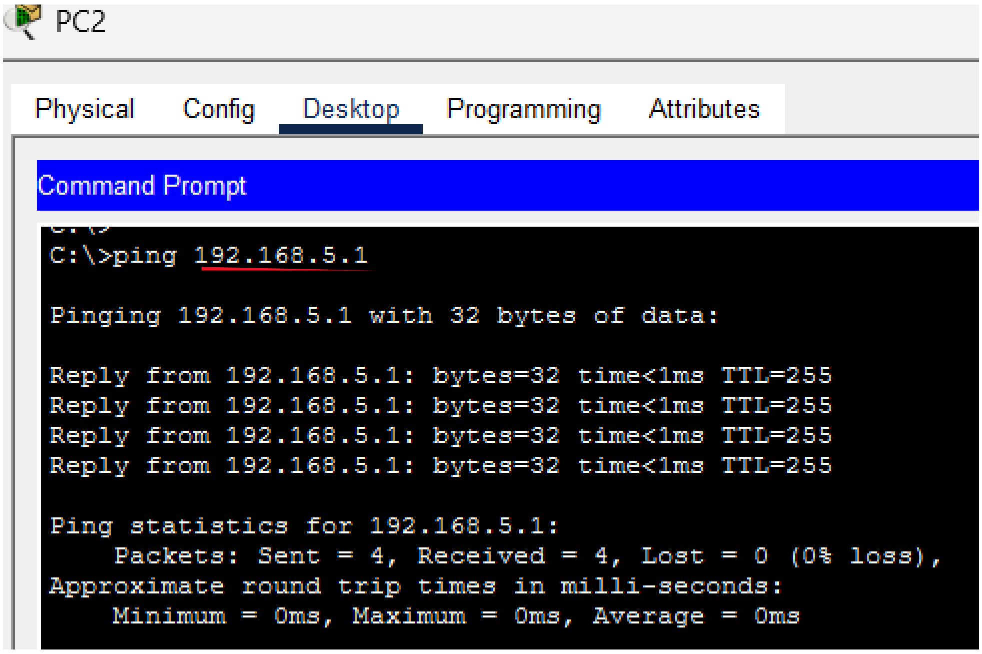




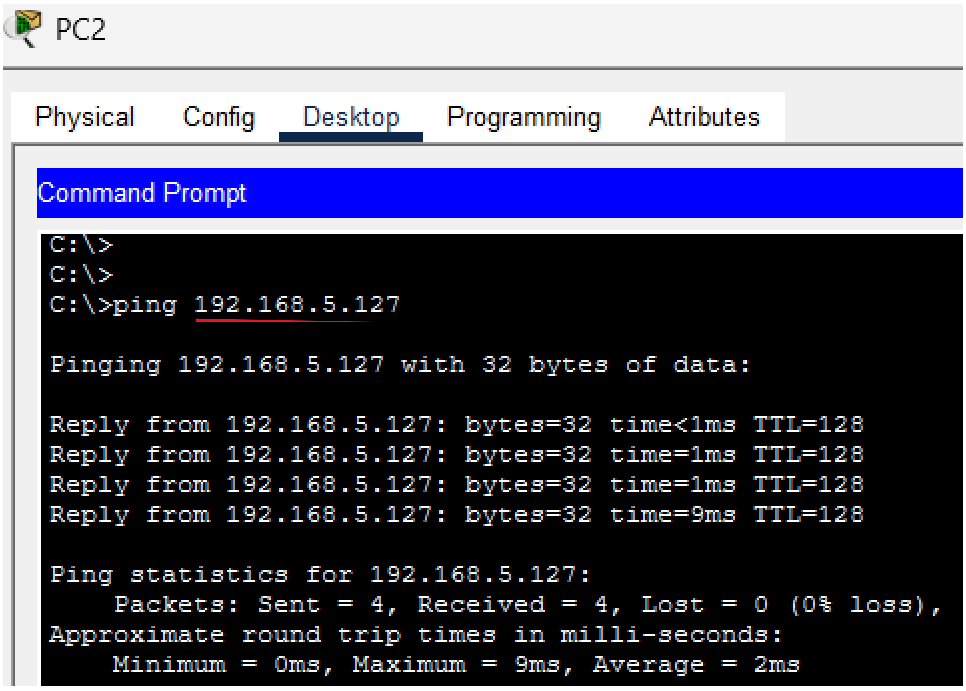
Part 6: Verify Connectivity

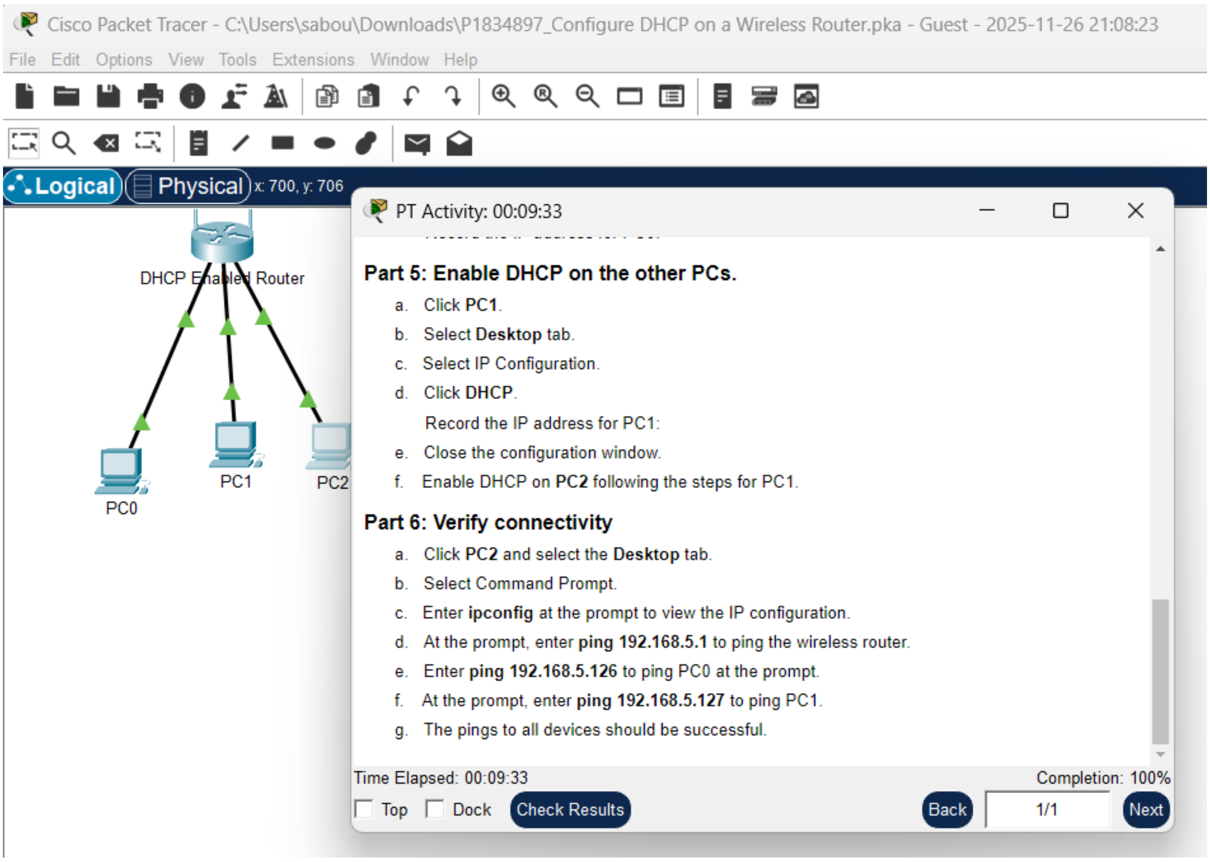
Ping between machines to validate network communication.

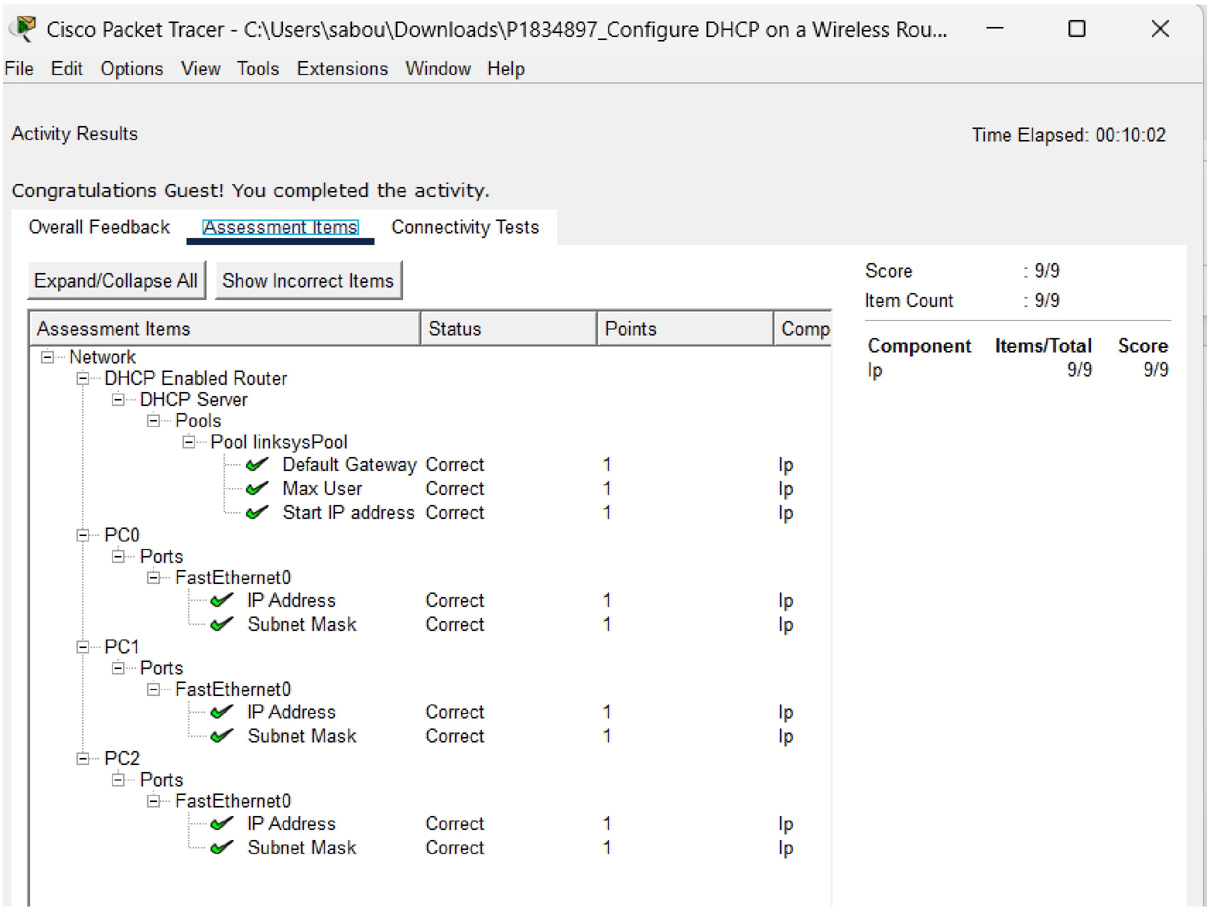
From PC2 I pinged **192.168.5.1** to ping the wireless router.



From PC2 I ping **192.168.5.127** to ping PC1




Cisco Packet Tracer - C:\Users\sabou\Downloads\P1834897_Configure DHCP on a Wireless Router.pka - Guest - 2025-11-26 21:08:23
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 700, y: 706
DHCP Enabled Router
PC0
PC1
PC2
PT Activity: 00:09:33
Part 5: Enable DHCP on the other PCs.
a. Click PC1.
b. Select Desktop tab.
c. Select IP Configuration.
d. Click DHCP.
Record the IP address for PC1:
e. Close the configuration window.
f. Enable DHCP on PC2 following the steps for PC1.
Part 6: Verify connectivity
a. Click PC2 and select the Desktop tab.
b. Select Command Prompt.
c. Enter ipconfig at the prompt to view the IP configuration.
d. At the prompt, enter ping 192.168.5.1 to ping the wireless router.
e. Enter ping 192.168.5.126 to ping PC0 at the prompt.
f. At the prompt, enter ping 192.168.5.127 to ping PC1.
g. The pings to all devices should be successful.
Time Elapsed: 00:09:33
Completion: 100%
Top
Dock
Check Results
Back
1/1
Next


Cisco Packet Tracer - C:\Users\sabou\Downloads\P1834897_Configure DHCP on a Wireless Rou...
File Edit Options View Tools Extensions Window Help
Activity Results
Time Elapsed: 00:10:02
Congratulations Guest! You completed the activity.
Overall Feedback
Assessment Items
Connectivity Tests
Expand/Collapse All
Show Incorrect Items
Assessment Items Status Points Comp
Network
DHCP Enabled Router
DHCP Server
Pools
Pool linksysPool
Default Gateway Correct 1 Ip
Max User Correct 1 Ip
Start IP address Correct 1 Ip
PC0
Ports
FastEthernet0
IP Address Correct 1 Ip
Subnet Mask Correct 1 Ip
PC1
Ports
FastEthernet0
IP Address Correct 1 Ip
Subnet Mask Correct 1 Ip
PC2
Ports
FastEthernet0
IP Address Correct 1 Ip
Subnet Mask Correct 1 Ip
Score : 9/9
Item Count : 9/9
Component Items/Total Score
Ip 9/9 9/9

Personal reflection

This activity allowed me to discover the role of the DHCP protocol. Thanks to this activity and the manipulations I performed, I understood how a DHCP server works. Configuring a wireless router also allowed me to strengthen my technical skills in all aspects of network management.